MATH3881/5881: Statistical Machine Learning Theory

Week 1: Tutorial Solutions — Introduction to Machine Learning

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Worked solutions to the Week 1 tutorial problems. Problem numbers and parts match the question sheet.

Problem 1: From ERM to the least-squares estimates

(a) C is a smooth convex quadratic in θ_0. Differentiating,

$$C'(\theta_0) \;=\; \frac{2}{n}\sum_{i=1}^{n}(\theta_0 - y^{(i)}) \;=\; \frac{2}{n}\Big(n\theta_0 - \sum_i y^{(i)}\Big).$$

Setting $C'(\theta_0) = 0$ gives the least-squares estimate $\hat\theta_0 = \frac{1}{n}\sum_i y^{(i)} = \bar y$. Since $C''(\theta_0) = 2 > 0$, this is the unique global minimum. ✓

(b) Differentiating $C(\theta_0, \theta_1) = \frac{1}{n}\sum_i (y^{(i)} - \theta_0 - \theta_1 x^{(i)})^2$:

$$\frac{\partial C}{\partial \theta_0} \;=\; -\frac{2}{n}\sum_i (y^{(i)} - \theta_0 - \theta_1 x^{(i)}) \;=\; 0 \;\;\Longrightarrow\;\; \bar y \;=\; \theta_0 + \theta_1 \bar x,$$

giving $\hat\theta_0 = \bar y - \theta_1 \bar x$. Substituting this into the other equation,

$$\frac{\partial C}{\partial \theta_1} \;=\; -\frac{2}{n}\sum_i x^{(i)}\big(y^{(i)} - \theta_0 - \theta_1 x^{(i)}\big) \;=\; 0,$$

and using $\theta_0 = \bar y - \theta_1 \bar x$:

$$\sum_i x^{(i)}\big(y^{(i)} - \bar y - \theta_1(x^{(i)} - \bar x)\big) \;=\; 0 \;\;\Longrightarrow\;\; \hat\theta_1 \;=\; \frac{\sum_i x^{(i)}(y^{(i)} - \bar y)}{\sum_i x^{(i)}(x^{(i)} - \bar x)}.$$

Finally, $\sum_i \bar x(y^{(i)} - \bar y) = 0$ and $\sum_i \bar x(x^{(i)} - \bar x) = 0$, so we may replace each $x^{(i)}$ in the numerator and denominator by $(x^{(i)} - \bar x)$:

$$\hat\theta_1 \;=\; \frac{\sum_i (x^{(i)} - \bar x)(y^{(i)} - \bar y)}{\sum_i (x^{(i)} - \bar x)^2}, \qquad \hat\theta_0 \;=\; \bar y - \hat\theta_1 \bar x. \quad ✓$$

(c) The first normal equation from part (b),

$$\frac{\partial C}{\partial \theta_0} \;=\; -\frac{2}{n}\sum_{i=1}^{n}\big(y^{(i)} - \hat\theta_0 - \hat\theta_1 x^{(i)}\big) \;=\; 0,$$

is exactly $\sum_i r^{(i)} = 0$ after substituting the fitted values $\hat\theta_0, \hat\theta_1$. Dividing through by n:

$$\bar y - \hat\theta_0 - \hat\theta_1\,\bar x \;=\; 0 \quad\Longrightarrow\quad \bar y \;=\; \hat\theta_0 + \hat\theta_1\,\bar x.$$

So the point $(\bar x, \bar y)$ lies on the fitted line — the least-squares fit always passes through the data's centre of mass. ✓

Problem 2: Probabilities and the decision boundary

(a) Let $z = -1 + 2x_1 - x_2$.

- $x = (1,1)$: $z = -1 + 2 - 1 = 0$, so $\hat y = \sigma(0) = 0.5$.
- $x = (2,1)$: $z = -1 + 4 - 1 = 2$, so $\hat y = \sigma(2) = \dfrac{1}{1+e^{-2}} \approx 0.881$.

(b) Using the hint, $\hat y > \tau \iff z > \log\frac{\tau}{1-\tau}$. Here $\log\frac{0.5}{0.5} = 0$ and $\log\frac{0.3}{0.7} \approx -0.847$.

x	z	$\tau = 0.5$ $(z > 0?)$	$\tau = 0.3$ $(z > -0.847?)$
$(1,1)$	0	no $\Rightarrow$ class 0	yes $\Rightarrow$ class 1
$(2,1)$	2	yes $\Rightarrow$ class 1	yes $\Rightarrow$ class 1

(At $\tau = 0.5$ the point $(1,1)$ sits exactly on the boundary; with the strict rule $\hat y > \tau$ it is classified 0.)

(c) Since σ is strictly increasing,

$$\hat y = \tau \iff \sigma\big(\hat\theta_0 + \hat\theta_1 x_1 + \hat\theta_2 x_2\big) = \tau \iff \hat\theta_0 + \hat\theta_1 x_1 + \hat\theta_2 x_2 \;=\; \log\frac{\tau}{1-\tau}.$$

Substituting $\hat\theta = (-1, 2, -1)$ and rearranging,

$$2x_1 - x_2 \;=\; 1 + \log\frac{\tau}{1-\tau}.$$

For $\tau = 0.5$: $2x_1 - x_2 = 1$. For $\tau = 0.3$: $2x_1 - x_2 = 1 + (-0.847) = 0.153$. Only the right-hand side (a constant) depends on τ; the coefficients of x_1, x_2 are fixed, so all such boundaries share the same slope — they are mutually *parallel*. Decreasing τ lowers the threshold $\log\frac{\tau}{1-\tau}$, so the half-plane classified as positive grows, and more points are predicted as class 1 (higher recall, lower precision, as discussed under Evaluation Metrics in the notes).

Problem 3: One step of gradient descent by hand

The design matrix (rows $\tilde x^{(i)\top} = (1, x^{(i)})$) and label vector are

$$X \;=\; \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}, \qquad y \;=\; \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad n = 2.$$

(a) At $\theta^{(0)} = (0,0)^\top$ we have $X\theta^{(0)} = (0,0)^\top$, so $\hat{y}^{(0)} = \sigma\big((0,0)\big) = (0.5,\, 0.5)^\top$. Initial cost (binary cross-entropy):

$$C(\theta^{(0)}) \;=\; -\frac{1}{2}\big[\,1\cdot\log 0.5 + 1\cdot\log 0.5\,\big] \;=\; -\log 0.5 \;=\; \log 2 \;\approx\; 0.693.$$

(b) Residual $\hat{y}^{(0)} - y = (0.5-1,\, 0.5-0)^\top = (-0.5,\, 0.5)^\top$. Then

$$X^\top(\hat{y}^{(0)} - y) \;=\; \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} -0.5 \\ 0.5 \end{bmatrix} \;=\; \begin{bmatrix} 0 \\ -1 \end{bmatrix}, \qquad \nabla_\theta C(\theta^{(0)}) \;=\; \frac{1}{2}\begin{bmatrix} 0 \\ -1 \end{bmatrix} \;=\; \begin{bmatrix} 0 \\ -0.5 \end{bmatrix}.$$

Update: $\theta^{(1)} \;=\; \theta^{(0)} - \alpha\,\nabla_\theta C(\theta^{(0)}) \;=\; (0,0)^\top - (0,-0.5)^\top \;=\; (0,\, 0.5)^\top$.

(c) Recall the gradient $\nabla_\theta C(\theta^{(0)}) = (0,\, -0.5)^\top$. The gradient points in the direction of steepest *increase* of C, so to reduce C we should move *opposite* to each component.

- **Intercept θ_0: hold fixed.** The 0-th component is zero, so C is (to first order) flat in the θ_0 direction at the current point — no motion is called for.
- **Slope θ_1: increase.** The 1-st component is negative (-0.5), so to decrease C we step in the *positive* θ_1 direction.

This agrees with the update from part (b): $\theta^{(1)} = (0,\, 0.5)^\top$ — θ_0 is unchanged and θ_1 has moved from 0 to $+0.5$.